

January 22, 2026

ECEN-607: Advanced Analog IC Design

Assignment #1: Transistor Characterization

Due: January 29, 2026

Homework will not be received after due.

Upload your solutions through Canvas

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Use the technology you are using in the lab and characterize a single N-type transistor employing $W=4.8\mu\text{m}$ and $L=0.18\mu\text{m}$. Use the IBM 0.18 technology you are using in the laboratory; please check it with your TA.

- i) Set the drain-source voltage at 1V, and sweep the gate-source voltage from 0 up to 1.5V. Plot i_{ds} , G_m , dG_m/dV_{gs} , and d^2G_m/dV_{gs}^2 versus gate-source voltage. Have enough points (>20) to have smooth curves.
- ii) For same conditions as i), plot C_{gs} and G_m/C_{gs} versus $V_{gs}-V_T$ (overdrive voltage); the parameter represents the transistor's f_t , which is quite relevant when designing fast circuits; G_m/C_{gs} is also known as the transistor's f_t .
- iii) Set the gate voltage such that the overdrive voltage $V_{gs}-V_T=200\text{mV}$, and sweep the drain voltage in the range 0-1.8V. Extract the transistor's output resistance and plot R_{ds} , dR_{ds}/dV_{ds} , and d^2R_{ds}/dV_{ds}^2 versus drain-source voltage. Notice that the derivatives of R_{ds} represent output resistor non-linearities. Certainly, R_{ds} is a second order effect and very non-linear, though!
- iv) Repeat iii) for temperatures of -50, 27 and 100 degrees (Celsius). Make a table and compare I_d , G_m and R_{ds} for the 3 cases. For all these simulations keep constant the gate-source voltage; evidently V_{dsat} will not be constant due to V_T variations.
- v) Simulate the transistor and report the input and output referred noise density. Include a screen shot showing the noise density in the frequency range of 1Hz up to 10MHz; use log10 scale in the x-axis and y-axis to visualize the flicker noise level (voltage) and the thermal noise level (current). Check these values with the theoretical values.

The easiest way to do this is by biasing the gate voltage with a DC voltage source such that $V_{gs}-V_T=200\text{mV}$ in series with an AC voltage source.

Connect at amplifier's drain terminal a power supply such that V_{DS} is set at 1V. The DC voltage source operates as short circuit for AC analysis and current can be measured.

Add conclusions in your report.